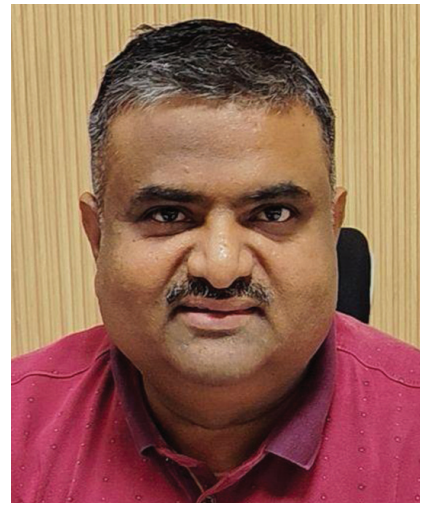
own non-contact pyro sensor with RS485 communication. This pattern of finding a practical problem and engineering a custom solution has become the company's defining approach to product development.

Alongside its own products, RHIA also undertakes specialised engineering projects. In one recent project, they delivered around 2000 printed circuit boards (PCBs) for an industrial application in the oil sector involving DC motor-driver control.

The broader direction reflects a shift towards automation systems that combine control, sensing, communication and connectivity more closely. For RHIA, the challenge is not simply to add more components to industrial machines, but to explore whether some of the technologies traditionally deployed separately can be integrated more effectively.

Founded in 2023 by Sharan Hiremath, the company operates on a bootstrapped model. The hardware and embedded development are



Sharan Hiremath, Founder and CEO, RHIA

carried out internally by RHIA's team, which also handles assembly and testing. The company's engineering team currently includes around 12 people working across hardware design, embedded firmware and software development.

On the thought behind the company's name, Praveen explains, "We chose the name RHIA because we wanted to represent our role as a helping hand for people and organisations that need support with hardware, embedded systems, IoT and industrial IoT technologies. RHIA stands for 'Right Hand Interactive Assistant'."

According to Sharan, the future of industrial automation is not about adding more components, but about making machines simpler, smarter, and more connected. "With RHIA," he says, "we are building that future from India."

While the company's journey is still at an early stage, its focus remains clear. They want to build reliable industrial technology from India for real-world industrial applications and create solutions that deliver measurable value to customers. This will be done by bringing control, sensing, connectivity, and intelligence together in a way that helps shape the next generation of industrial machines and automation systems.

—Ananthu Ashok

2 VIMAG LABS' MAGNET-FREE MOTORS SET TO REDEFINE EV EFFICIENCY

Can software replace rare-earth materials? This Bengaluru startup uses virtual magnets to eliminate traditional permanent magnets, boosting EV motor efficiency up to 95 per cent.

Imagine building a high-performance electric vehicle (EV) motor without using a single rare-earth permanent magnet. By deploying software-controlled electronics to generate a 'Virtual Magnet', Vimag Labs, a Bengaluru-based deeptech startup, builds lighter, highly efficient systems tailored to address the entire gamut of EVs.

Founded by Manish Seth and Dr Piyush Desai in 2025, Vimag Labs utilises software-controlled electronics to generate internal magnetic fields, eliminating the permanent magnets traditionally required. This patented 'Virtual Magnet Technology' is engineered to match or exceed the performance of conventional



Vimag Labs' Bengaluru R&D and testing facility

permanent-magnet systems.

The technology can be used across all EV segments, drones, robotics, actuators, and industrial equipment. Vimag Labs also develops the electronics, software, and control systems required to generate and manage the virtual magnetic field inside the motor.

"Our technology uses wireless